Ioan-Teodor Moroșanu

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01 WORK EXPERIENCE

Summer Intern
Analog Devices Romania
07/2026 – 08/2026 Cluj-Napoca, Romania

Designed a custom FPGA-compatible PCB with an onboard accelerometer that triggers dynamic LED indicators. During this time, I gained hands-on experience in:

- **Hardware design**:
  - Simulated analog circuits in **LTspice.**
  - Designed schematics and PCBs using **KiCad.**
- **FPGA & HDL Development:**
  - Created custom IP core in **Verilog.**
  - Implemented FPGA design using **Xilinx Vivado.**
- **Software design**:
  - Developed embedded scripts in **Visual Studio Code.**
  - Streamlined CI/CD workflows using **GitHub Actions.**
  - Built application scripts using **Python.**
  - Conducted most software development within a **Linux-based environment**.

AI Model Evaluator & Data Contributor
Prolific Academic
02/2026 – present Remote, Romania

Actively participating in specialized AI training and academic research studies, focusing on **QA** and **RLHF**:

- **Technical & CAD Evaluation:** Reviewed AI-generated Python code and tested generative CAD models to identify syntax errors and design limitations.
- **Language & Multimodal QA:** Evaluated Romanian speech synthesis, images, and videos by comparing outputs for naturalness, accuracy, and prompt adherence.
- **Research Contributions:** Provided structured analytical feedback and data validation across interdisciplinary academic and technical studies.

02 EDUCATION AND TRAINING

Automation, applied informatics and smart systems | Bachelor's Degree
Technical University of Cluj-Napoca, Faculty of Automation and Computer Science
10/2023 – present Cluj-Napoca, Romania

03 HONOURS AND AWARDS

1st Prize - ROSPIN School (national competition)
ROMANIAN SPACE INITIATIVE
05/2022
https://m.monitorulsv.ro/Ultima-ora-local/2022-05-24/Elevi-de-la-Colegiul-Alexandru-cel-Bun-Gura-Humorului-si-Colegiul-Dragos-Voda-Campulung-Moldovenesc-castigatori-ai-unei-competitii-nationale-de-rovere?fbclid=IwAR1rqj5rgaVZkXN2ncqrQUSqgzJg7bpsEUwak6b7LIGPC44ZDtKEXTejTPU

Designing a rover for a **moon mission** that meets the specific criteria of a lunar vehicle, featuring innovative design elements.

04 PERSONAL PROJECTS

Hardware Projects:

- **Train Network (1:50 Scale):** A 3D-printed model railway featuring ESP32-powered locomotives, RFID and gyroscope-based digital track mapping, Wi-Fi captive portal, OTA firmware updates, and a Unity control app.
- **Sunflower (Autonomous Self-Watering Vase):** An ESP32 and solar-powered smart vase featuring capacitive soil moisture sensing, automated pump control, ambient/grow LEDs, and real-time Wi-Fi monitoring via a Unity mobile app.
- **Motion Box (Remote Security System):** A wireless intrusion detection system using a PIR sensor, long-range NRF transceivers for remote buzzer/LED alerts, and an integrated camera for automated photo uploads to Google Drive.
- **8.4V 2S 18650 Battery Charger:** A compact power management unit with USB Type-C power delivery, 8.4V charging circuitry for 2S lithium-ion packs, LED status indicators, and a custom 3D-printed enclosure.
- **Repository:** https://github.com/teodormorosanu/Electronic-Projects

Software Projects:

- **Zombie-Game (3D Desktop FPS):** A first-person parkour shooter prototype utilizing Unity's Universal Render Pipeline (URP) and custom shaders, featuring physics-driven traversal, a real-time minimap, and a locational hitbox system for dynamic enemy dismemberment.
- **Subdive (3D Mobile Underwater Exploration):** A mobile submarine simulator featuring custom underwater buoyancy physics, aquatic shaders, spatial audio, and a modular progression system for vessel upgrades and in-game economy.
- **DirectMaze (Mobile Algorithmic Puzzle):** A matrix-based logic puzzle game centered on spatial reasoning and finite-move optimization, challenging players to compute optimal pathfinding routes to collect all objectives under strict movement constraints.
- **CusCus – PoliHack v15 (Educational Electronics App):** An interactive gamified application designed to teach fundamental electronics concepts to children through intuitive visual simulations.
- **Repositories:** https://github.com/teodormorosanu/Zombie-Game | https://github.com/teodormorosanu/Subdive | https://github.com/teodormorosanu/DirectMaze | https://github.com/teodormorosanu/CusCusPolihackv15

YouTube Channels:

- **teoelectric:** Engineering and maker channel documenting the step-by-step design, 3D printing, electronics assembly, and testing of the 1:50 scale model train network.
- **T3do Dev:** Game development channel featuring devlogs, gameplay trailers, and technical updates on custom interactive applications and games built in Unity.
- **Romanian SR Beats:** Music production and audio editing channel focused on remixing tracks through tempo modulation and spatial audio effects (slowed + reverb).
- **Links:** https://www.youtube.com/@teoelectric | https://www.youtube.com/@t3dodev | https://www.youtube.com/@rosrbeats

05 SKILLS

PROGRAMMING AND SOFTWARE DEVELOPMENT

C C++ C# Python MATLAB Java Unity Git GitHub Actions

HARDWARE, EMBEDDED SYSTEMS AND CAD

Espressif STM Verilog KiCad LTSpice Xilinx Vivado Linux Fusion360 AutoCAD

DATABASES AND PACKAGES

MySQL SQL Server Microsoft Office Adobe Creative Cloud

LANGUAGES

Romanian Native **English** B2 **Spanish** A1 **Polish** A1